REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Matter Waves Practice Questions

Grade 12 Advanced | Module 22 Lesson 2 | Matter Waves

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: $\lambda = h/p = h/(mv)$ for nonrelativistic particles; $\Delta x \Delta p \geq h/(4\pi)$. $h = 6.626 \times 10^{-34}$ J s; $m_e = 9.11 \times 10^{-31}$ kg.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 59–61.

P01. The de Broglie wavelength of a particle is:

A. hp B. p/h
C. h/p D. mc^2

P02. If particle momentum doubles, its de Broglie wavelength:

A. Doubles B. Quadruples
C. Stays unchanged D. Halves

P03. Electron diffraction demonstrates electrons' :

A. Wave behaviour B. Zero momentum
C. Lack of mass D. Classical paths only

P04. The uncertainty principle concerns a fundamental limit on simultaneous knowledge of:

A. Mass and charge only B. Position and momentum
C. Wavelength and colour only D. Energy and mass in every measurement

---

P05. Macroscopic objects normally show negligible diffraction because their de Broglie wavelengths are:

A. Larger than the room B. Always exactly zero
C. Extremely small compared with relevant openings D. Independent of momentum

---

P06. A particle has momentum 2×10^{-24} kg m/s. Find its de Broglie wavelength. Use $h = 6.626 \times 10^{-34}$ J s.

A. 3.31×10^{-10} m B. 6.63×10^{-10} m
C. 1.66×10^{-10} m D. -3.31×10^{-10} m

P07. An electron moves at 2×10^{6} m/s. Find its nonrelativistic de Broglie wavelength. Use $m_e = 9.11 \times 10^{-31}$ kg and $h = 6.626 \times 10^{-34}$ J s.

A. 7.27×10^{-10} m B. 3.64×10^{-10} m
C. 1.82×10^{-10} m D. -3.64×10^{-10} m

P08. A particle has de Broglie wavelength 2×10^{-10} m. Find its momentum. Use $h = 6.626 \times 10^{-34}$ J s.

A. 6.63×10^{-24} kg m/s B. 1.66×10^{-24} kg m/s
C. 3.31×10^{-24} kg m/s D. -3.31×10^{-24} kg m/s

P09. The momentum of a particle is multiplied by 3. Calculate λnew/λold for its de Broglie wavelength.

A. 0.6667 B. 0.1667
C. -0.3333 D. 0.3333

P10. Use $\Delta x \Delta p \geq h/(4\pi)$ to find the minimum momentum uncertainty for $\Delta x = 2 \times 10^{-10}$ m. Use $h = 6.626 \times 10^{-34}$ J s.

A. 2.64×10^{-25} kg m/s B. 5.27×10^{-25} kg m/s
C. 1.32×10^{-25} kg m/s D. -2.64×10^{-25} kg m/s

P11. State de Broglie's hypothesis.

P12. Why is electron diffraction evidence for matter waves?

P13. State Heisenberg's position–momentum uncertainty principle.

P14. How does increasing speed affect the de Broglie wavelength of the same nonrelativistic particle?

P15. Does the uncertainty principle mean only that instruments are poorly made?

P16. A particle has momentum 3×10^{-24} kg m/s. Find its de Broglie wavelength. Use $h = 6.626 \times 10^{-34}$ J s.

P17. A particle has momentum 4×10^{-24} kg m/s. Find its de Broglie wavelength. Use $h = 6.626 \times 10^{-34}$ J s.

P18. An electron moves at 3×10^6 m/s. Find its nonrelativistic de Broglie wavelength. Use me = 9.11×10^{-31} kg and h = 6.626×10^{-34} J s.

P19. An electron moves at 4×10^6 m/s. Find its nonrelativistic de Broglie wavelength. Use me = 9.11×10^{-31} kg and h = 6.626×10^{-34} J s.

P20. A particle has de Broglie wavelength 3×10^{-10} m. Find its momentum. Use h = 6.626×10^{-34} J s.

P21. A particle has de Broglie wavelength 4×10^{-10} m. Find its momentum. Use h = 6.626×10^{-34} J s.

P22. The momentum of a particle is multiplied by 4. Calculate $\lambda_{new}/\lambda_{old}$ for its de Broglie wavelength.

P23. The momentum of a particle is multiplied by 5. Calculate $\lambda_{new}/\lambda_{old}$ for its de Broglie wavelength.

P24. Use $\Delta x \Delta p \geq h/(4\pi)$ to find the minimum momentum uncertainty for $\Delta x = 3 \times 10^{-10}$ m. Use h = 6.626×10^{-34} J s.

P25. Use $\Delta x \Delta p \geq h/(4\pi)$ to find the minimum momentum uncertainty for $\Delta x = 4 \times 10^{-10}$ m. Use $h = 6.626 \times 10^{-34}$ J s.

P26. An electron and a proton have equal momentum. Compare their de Broglie wavelengths.

P27. An electron and a proton have equal nonrelativistic speeds. Which has the longer wavelength?

P28. If position uncertainty is reduced to one quarter, how does the minimum momentum uncertainty change?

P29. Why can a crystal diffract electrons?

P30. Explain why treating an electron as a tiny ball on a precisely known trajectory is limited.